Capacitated Vehicle Routing Quantum Clustering Optimization Using Analog Neutral Atoms with Compact Slack-Free Encoding

Aiden C Khanna, Yousif Ashour, Isaac Atteson

DuQuantum 2026 - Pasqal Track A

October 25, 2026

Abstract

We investigate a compact, slack-free QUBO formulation for capacity-constrained clustering within a cluster-first, route-second approach to the Capacitated Vehicle Routing Problem. For four customers and two vehicles, the formulation uses four binary variables, reducing the atom requirement by 71.4% relative to one-hot assignment with binary capacity slack. Capacity constraints are approximated through a quadratic load-balancing penalty, and all measured assignments are checked independently for feasibility. We implement the model through analog neutral-atom emulation with global detuning and evaluate 20 certified instances against exact classical references and uniform random sampling. Emulation achieves mean feasibility and clustering-optimum probabilities of 96.47% and 63.97%, compared with 58.66% and 13.10% for random sampling. Performance varies substantially: median clustering success is 90.85% when every embedded ground state is clustering-optimal and 2.45% otherwise. Exact routing of measured clusters recovers the CVRP optimum on 17 of 20 instances, demonstrating the distinction between clustering and routing quality. A preliminary 20-atom extension produces only 0.5% feasible measurements despite satisfying geometric limits. These results demonstrate compact encoding and improved mean sampling probabilities on small instances, while identifying Hamiltonian fidelity and constraint enforcement as barriers to scaling. They do not establish a computational advantage.

1. Challenge Statement and Context

1.1. Capacitated Vehicle Routing

The Pasqal challenge at DuQuantum 2026 concerned the Capacitated Vehicle Routing Problem (CVRP): assigning customers to a fixed fleet of identical vehicles and determining their visiting order while minimizing total travel distance [1]. Each customer has a known location and demand. Every customer must be served exactly once, and the total demand assigned to a vehicle must not exceed its capacity.

The challenge assumes a fixed depot, symmetric travel costs satisfying the triangle inequality, and unsplit deliveries. Each used vehicle makes one trip, beginning and ending at the depot without returning to reload. Vehicles may remain unused, in which case they contribute no travel cost. Although no individual demand exceeds vehicle capacity, these assumptions alone do not guarantee that a feasible assignment exists.

1.2. Decomposition and Scope

To reduce the encoding requirements, the challenge adopts a *cluster-first, route-second* decomposition. The first stage assigns customers to capacity-feasible groups, with each group served by one vehicle. The second stage determines the visiting order within each group, forming a Traveling Salesman Problem that includes the depot.

The challenge offers two tracks. Track A, addressed in this work, requires capacity-feasible clustering using a participant-defined measure of geographic proximity. Track B additionally requires quantum optimization of the routes within the resulting clusters. Total pairwise distance, distance to a cluster center, and other clustering objectives are permitted; selecting and justifying the objective is part of the challenge.

This decomposition is heuristic. A partition minimizing a clustering objective need not produce the shortest routes, even when each cluster is subsequently routed exactly. Accordingly, this work evaluates both the chosen clustering objective and the total route length obtained through exact classical routing of the measured clusters. The latter evaluates the downstream quality of the clustering without introducing a quantum routing stage.

1.3. Instance Requirements

The challenge specifies three instance tiers [1]:

Tier	Customers (N)	Vehicles (K)	Capacity (Q)	Customer demand range
S	4	2	7	$\{1, \ldots, 4\}$
M	10	3	20	$\{1, \ldots, 8\}$
L	15	4	30	$\{1, \ldots, 10\}$

Customer and depot locations lie within a square of side 100 distance units. Within these restrictions, participants select the depot position, customer locations, individual demands, and random seeds.

Instances must be reproducible and certified feasible. Each instance therefore requires at least one valid customer-to-vehicle assignment that can be checked independently. The challenge also requires reporting the load ratio and largest-demand ratio:

$$\rho = \frac{\sum_i d_i}{KQ}, \quad \frac{d_{\max}}{Q}, \qquad d_{\max} = \max_i d_i. \tag{1}$$

These quantities characterize aggregate capacity utilization and the size of individual demands relative to vehicle capacity.

1.4. Neutral-Atom Implementation Requirements

The central objective is to encode the constrained problem using few binary variables while retaining interactions that can be implemented on an analog neutral-atom register. In a direct encoding, each binary variable requires one atom. Auxiliary variables therefore consume register capacity and introduce additional interactions that must be embedded.

The challenge specifies the following hardware envelope, based on Pulser's *AnalogDevice* object definition [1]:

Parameter	Challenge specification
Maximum atom count	80
Minimum interatomic separation	$5, \mu\mathrm{m}$
Maximum register radius	$38, \mu\mathrm{m}$
Maximum Rabi frequency	$12.57, \mathrm{rad}/\mu\mathrm{s}$
Detuning range	$\pm 125.66, \mathrm{rad}/\mu\mathrm{s}$
Maximum sequence duration	$6000, \mathrm{ns}$
Interaction coefficient $C_6/\hbar$	$8.657 \times 10^5, (\mathrm{rad}/\mu\mathrm{s}), \mu\mathrm{m}^6$

Interactions are repulsive and decay with the sixth power of interatomic distance. Consequently, arbitrary coupling matrices cannot be reproduced simply by choosing atom positions. Global detuning also applies the same linear coefficient to every atom, restricting the objectives that can be represented exactly.

Our experiments used emulators rather than physical quantum hardware. Detuning map modulation (DMM), allowing restricted site-dependent detuning, was permitted in emulation, but its results had to be labeled and compared separately from global-detuning results, therefore were to avoided if possible.

1.5. Evaluation Criteria

Track A requires evaluating two distinct solution-quality gaps. The *model gap* compares the clustering objective of the best feasible measured assignment with the optimum of the original constrained clustering model. The *CVRP gap* compares the total route length obtained by routing the measured clusters exactly with a reference for the full CVRP.

For either objective, the percentage gap is

$$\mathrm{Gap}(\%) = \frac{C - C^\star}{C^\star} \times 100\%, \tag{2}$$

where C is the feasible solution cost and $C^\star$ is the corresponding reference cost. A penalized QUBO optimum must not be assumed equivalent to the constrained-model optimum.

The required metrics also include the fraction of feasible measurements, the probability of reaching the reference within a stated tolerance, total atom count, and relative interaction error. Reference values must be identified as proven optima or best-known solutions. Assignments differing only by vehicle labels represent the same clustering.

2. Tier-S Model

2.1. Problem Definition

We consider $N = 4$ customers and $K = 2$ identical vehicles, each with capacity $Q = 7$. Customer i has position $\mathbf{r}_i$ and demand $d_i \in 1, 2, 3, 4$. Define the customer-to-customer distances and total demand as

$$D_{ij} = |\mathbf{r}_i - \mathbf{r}_j|_2, \qquad T = \sum_{i=1}^{N} d_i. \tag{3}$$

Each customer is represented by one binary assignment variable:

$$z_i = \begin{cases} 1 & \text{if assigned to vehicle 1} \\ 0 & \text{if assigned to vehicle 2.} \end{cases} \tag{4}$$

This encoding assigns every customer to exactly one vehicle without an additional assignment penalty. The vehicle loads are

$$L_1(z) = \sum_i d_i z_i, \qquad L_2(z) = T - L_1(z). \tag{5}$$

We choose to minimize the total pairwise distance between customers sharing a vehicle. The same-vehicle indicator is

$$\begin{aligned} &z_i z_j + (1 - z_i)(1 - z_j) \\ &= 1 - z_i - z_j + 2 z_i z_j. \end{aligned} \tag{6}$$

It equals one when $z_i = z_j$ and zero otherwise. The clustering objective is therefore [2]:

$$\boxed{C(z) = \sum_{i<j} D_{ij}(1 - z_i - z_j + 2 z_i z_j).} \tag{7}$$

Minimizing $C(z)$ discourages placing geographically distant customers in the same vehicle. It does not include depot distances or visiting order; these enter the subsequent routing evaluation.

The original constrained clustering problem is

$$\begin{aligned} \min_{z \in \{0,1\}^N} \quad & C(z) \\ \text{subject to} \quad & L_1(z) \le Q, \\ & L_2(z) \le Q. \end{aligned} \tag{8}$$

Equivalently, both capacity constraints hold precisely when

$$T - Q \le L_1(z) \le Q. \tag{9}$$

Because the vehicles are identical, z and $1 - z$ describe the same unlabeled partition.

2.2. Slack-Free Capacity Penalization

A conventional QUBO treatment introduces nonnegative slack variables to convert capacity inequalities into equalities, then penalizes squared deviations [3], [4], [5]. To avoid the additional binary variables, we instead use a quadratic surrogate motivated by unbalanced penalization [6].

Define the remaining capacity of each vehicle:

$$\begin{aligned} h_1(z) &= Q - L_1(z), \\ h_2(z) &= Q - L_2(z). \end{aligned} \tag{10}$$

Capacity feasibility requires $h_1(z) \ge 0$ and $h_2(z) \ge 0$. Applying common linear and quadratic weights gives

$$\begin{aligned} P_{\text{raw}}(z) = \; & -\alpha\Big(h_1(z) + h_2(z)\Big) \\ & + \lambda\Big(h_1(z)^2 + h_2(z)^2\Big), \\ & \alpha, \lambda > 0. \end{aligned} \tag{11}$$

For this two-vehicle encoding, the sum of the margins is constant:

$$h_1(z) + h_2(z) = 2Q - T. \tag{12}$$

Consequently, the linear term does not affect the minimizing assignment. The quadratic terms satisfy

$$\begin{aligned} h_1(z)^2 + h_2(z)^2 = \; & 2\left(L_1(z) - \frac{T}{2}\right)^2 \\ & + 2\left(Q - \frac{T}{2}\right)^2. \end{aligned} \tag{13}$$

Removing assignment-independent terms leaves

$$\boxed{P(z) = 2\lambda\left(\sum_i d_i z_i - \frac{T}{2}\right)^2.} \tag{14}$$

Thus, although motivated by unbalanced penalization, the resulting penalty is a symmetric load-balancing term. It favors loads near $T/2$ rather than penalizing only capacity violations.

The complete unconstrained objective is

$$\boxed{\begin{aligned} F(z) = \; & \sum_{i<j} D_{ij}(1 - z_i - z_j + 2 z_i z_j) \\ & + 2\lambda\left(\sum_i d_i z_i - \frac{T}{2}\right)^2. \end{aligned}} \tag{15}$$

This surrogate is related to capacity feasibility through

$$
\begin{aligned}
&L_1(z) \le Q,\ L_2(z) \le Q \\
&\Longleftrightarrow \\
&\left| L_1(z) - \frac{T}{2} \right| \le Q - \frac{T}{2}.
\end{aligned}
\tag{16}
$$

However, the penalty also distinguishes between feasible assignments with different load imbalances. It can therefore change which feasible clustering is preferred. Moreover, for fixed λ, Q appears only in discarded constants, so changing capacity alone does not change the unconstrained objective.

We use $\lambda = 0.1$ as a heuristic baseline. Increasing λ prioritizes demand balance, while decreasing it prioritizes geographic clustering. Its numerical significance depends on the distance and demand scales; it is not an instance-independent constraint-enforcement threshold.

Every measured assignment is checked against Equation (8). Only feasible assignments contribute to the reported solution-quality gaps, which use the original clustering cost $C(z)$ rather than the penalized objective $F(z)$.

2.3. QUBO Representation

Using the binary identity $z_i^2 = z_i$, the squared load expands as

$$
\begin{aligned}
\left(\sum_i d_i z_i \right)^2 &= \sum_i d_i^2 z_i \\
&\quad + 2 \sum_{i<j} d_i d_j z_i z_j.
\end{aligned}
\tag{17}
$$

Collecting the constant, linear, and quadratic terms gives the symmetric QUBO representation [3]:

$$
\boxed{F(z) = c + z^{\mathsf{T}} M z.}
\tag{18}
$$

The diagonal coefficients are

$$
M_{ii} = -\sum_{j \neq i} D_{ij} + 2\lambda(d_i^2 - T d_i).
\tag{19}
$$

and the off-diagonal coefficients are

$$
M_{ij} = D_{ij} + 2\lambda d_i d_j, \qquad i \neq j.
\tag{20}
$$

The constant offset is

$$
c = \sum_{i<j} D_{ij} + \frac{\lambda T^2}{2}.
\tag{21}
$$

Because M is symmetric,

$$
z^{\mathsf{T}} M z = \sum_i M_{ii} z_i + 2 \sum_{i<j} M_{ij} z_i z_j.
\tag{22}
$$

Accordingly, the coefficient of each pairwise product is $2M_{ij}$, not M_{ij}. In the alternative convention

$$
F(z) = c + \sum_i h_i z_i + \sum_{i<j} J_{ij} z_i z_j,
\tag{23}
$$

the coefficients are $h_i = M_{ii}$ and $J_{ij} = 2M_{ij}$. This distinction is required when matching the QUBO to physical interactions.

The offset c can be omitted during optimization without changing the minimizers. All off-diagonal entries are positive for positive demands and $\lambda > 0$, matching the sign of repulsive Rydberg interactions. This sign compatibility does not guarantee geometric embeddability. The unequal diagonal entries also prevent exact direct matching with a single global detuning [7].

2.4. Variable Efficiency

A conventional one-hot assignment encoding uses $NK = 8$ variables. Representing each vehicle's capacity slack from zero through seven requires three additional bits per vehicle, giving 14 variables in total [3].

The compact assignment encoding can also be combined with conventional capacity slack. Including this baseline separates the savings from assignment encoding from those obtained by removing slack.

Encoding	Assignment bits	Slack bits	Total
One-hot assignment only	8	0	8
One-hot assignment with binary slack	8	6	14
Compact assignment with binary slack	4	6	10
Proposed compact balancing surrogate	4	0	**4**

The proposed formulation therefore uses

$$
\boxed{N_{\text{bits}} = N = 4.}
\tag{24}
$$

This is a reduction of 71.4% relative to the conventional one-hot formulation with slack, or 60% relative to compact assignment with slack. Under the direct neutral-atom mapping, these four variables require four atoms. The reduction comes at the cost of replacing exact capacity encoding with a load-balancing surrogate whose outputs require independent feasibility checks.

Variable Efficiency and Scalability

The conventional formulation introduces binary assignment variables x_{ik}, assignment penalties, and binary-encoded capacity slack s_k:

$$F_{\text{std}} = C(x) + A\sum_i \left(\sum_k x_{ik} - 1\right)^2 + B\sum_k \left(\sum_i d_i x_{ik} + s_k - Q\right)^2. \quad (25)$$

Our reduction eliminates the need for slack variables; the resulting variable requirements are compared below:

Encoding	Variable count	S	M	L	Fits in 80 atoms?
Assignment variables only	NK	8	30	60	Yes
+ Binary-encoded slack	$NK + K\lceil \log_2(Q+1)\rceil$	14	45	80	Yes
Our compact formulation	$N\lceil \log_2 K\rceil$	**4**	**20**	**30***	Yes*

The Tier L value is a projected assignment-variable count, not a fully implemented quadratic formulation. The conventional Tier L model reaches the 80-atom limit exactly.

Our approach reduces the Tier S variable requirement by 71.4% relative to the conventional slack-variable formulation. However, a smaller register does not guarantee that its interactions can be accurately embedded. This is apparent when we attempted to assess an M-Tier model by embedding a 20 qubit array, as we were unable to achieve a feasible layout using neutral-atom emulation. The original unbalanced penalization method is designed around quantum annealing, therefore scalability may be unfeasible analog neutral-atom systems, more work needs to be done.

3. Quantum Implementation and Emulation

3.1. Hamiltonian Mapping and Embedding

Each assignment variable is represented by one neutral atom, with $|0\rangle = |g\rangle$ and $|1\rangle = |r\rangle$. The excitation-number operator $\hat{n}_i = |r\rangle_i\langle r|$ therefore represents the binary variable z_i. For a global drive with laser phase zero, the register evolves under [7]

$$\hat{H}(t) = \frac{\Omega(t)}{2}\sum_i X_i - \delta(t)\sum_i \hat{n}_i + \sum_{i<j}\frac{\widetilde{C}_6}{R_{ij}^6}\hat{n}_i\hat{n}_j, \quad (26)$$

where $\Omega(t)$ is the Rabi frequency, $\delta(t)$ is the global detuning, $\widetilde{C}_6 = C_6/\hbar$, and R_{ij} is the interatomic separation. The Rabi term couples computational-basis configurations, while the detuning and interaction terms assign their energies.

At the end of the pulse, $\Omega(\tau) = 0$. Ignoring the constant QUBO offset, the desired final Hamiltonian is

$$\hat{H}_{\text{target}} = \gamma\sum_i M_{ii}\hat{n}_i + 2\gamma\sum_{i<j} M_{ij}\hat{n}_i\hat{n}_j, \quad (27)$$

where $\gamma > 0$ converts the objective coefficients into angular-frequency units. The factor of two follows from the symmetric QUBO convention.

Exact interaction matching would require

$$R_{ij}^\star = \left(\frac{\widetilde{C}_6}{2\gamma M_{ij}}\right)^{1/6}. \quad (28)$$

The atom coordinates are distinct from the customer coordinates. They encode the desired interactions: a larger positive QUBO coefficient requires a smaller atom separation.

The target distances need not admit an exact two-dimensional Euclidean embedding [8]. We therefore optimize atom positions using sequential least-squares programming (SLSQP), minimizing the sum of squared interaction mismatches subject to

$$R_{ij} \geq 5, \mu\text{m}, \qquad |\mathbf{a}_i|_2 \leq 38, \mu\text{m}, \quad (29)$$

where $\mathbf{a}_i$ is the position of atom i. The realized interactions are $V_{ij} = \widetilde{C}_6/R_{ij}^6$, and their relative error is

$$\epsilon_V = \sqrt{\frac{\sum_{i<j}(V_{ij} - V_{ij}^\star)^2}{\sum_{i<j}(V_{ij}^\star)^2}}. \quad (30)$$

This metric measures pair-interaction error. It does not include the approximation of the diagonal coefficients or guarantee preservation of the target ground states.

3.2. Global Detuning and Pulse Design

Exact diagonal matching would require individual detunings $\delta_i = -\gamma M_{ii}$. Because our implementation uses global detuning, the unequal diagonal coefficients are approximated by their mean:

$$\delta_f = -\frac{\gamma}{N}\sum_i M_{ii}. \tag{31}$$

Pasqal's QUBO solver also uses a mean-detuning approximation when DMM is disabled, with its own coefficient normalization [9]. For our convention, Equation (31) minimizes the sum of squared diagonal-coefficient errors at fixed γ.

Nevertheless, averaging changes the objective. Before geometric embedding error is introduced, the difference between the global-detuning energy and the target energy is

$$\Delta E(z) = \sum_i (-\delta_f - \gamma M_{ii}) z_i. \tag{32}$$

Consequently, a device-legal sequence need not have the same minimizing assignments as the original QUBO. We check ground-state preservation separately from interaction fidelity.

We use an adiabatic-inspired schedule based on the challenge baseline and Pulser's QUBO tutorial [1], [7]. The squared-sine amplitude envelope has precedent in Rydberg-array optimization [10] and provides zero amplitude and zero slope at both endpoints. The linear detuning sweep connects the initial negative detuning to the final value required by the encoded objective. These pulses are not fully adiabatic, therefore doesn't guarantee an optimal schedule:

$$\begin{aligned} \Omega(t) &= \Omega_p \sin^2\left(\frac{\pi t}{\tau}\right), \\ \delta(t) &= \delta_0 + (\delta_f - \delta_0)\frac{t}{\tau}, \\ 0 &\le t \le \tau. \end{aligned} \tag{33}$$

The pulse parameters are selected heuristically:

$$\begin{aligned} \Omega_p &= \mathrm{median}_{i<j}(2\gamma M_{ij}), \\ \delta_0 &= -\gamma \max_i |M_{ii}|, \\ \tau &= 4000\,\mathrm{ns}. \end{aligned} \tag{34}$$

The median interaction supplies a representative drive scale [5]. The common scale γ is restricted by the device's Rabi-frequency, detuning, and minimum-separation limits, using a 10% safety margin. The fixed $4,\mu$s duration lies within the challenge's $6,\mu$s limit.

Initially, $\Omega(0) = 0$ and $\delta_0 < 0$, so the all-ground-state register is the lowest-energy configuration. During the sweep, the Rabi drive allows transitions between configurations before returning to zero. These choices do not guarantee adiabatic evolution: finite-duration dynamics and errors in the final Hamiltonian can both affect the measured distribution.

3.3. Example Register and Emulation

Figure 1 shows the embedded register for the Tier-S instance with seed 1 and demands $\{3, 2, 2, 3\}$. The atom labels $q0, \ldots, q3$ correspond to assignment variables $z_1, \ldots, z_4$.

Parameter	Value
Energy scale γ	0.0996249
Peak Rabi frequency Ω_p	$11.3097, \mathrm{rad}/\mu\mathrm{s}$
Initial detuning δ_0	$-20.4782, \mathrm{rad}/\mu\mathrm{s}$
Final detuning δ_f	$17.7852, \mathrm{rad}/\mu\mathrm{s}$
Minimum atom separation	$6.0496, \mu\mathrm{m}$
Maximum distance from register origin	$5.4702, \mu\mathrm{m}$
Relative interaction error ϵ_V	23.49%

The diagram uses the characteristic blockade radius evaluated at the peak drive:

$$6.5161, \mu\mathrm{m}.$$

Each circle shows the **full radius** R_b. A pair lies within this scale when one atom's center lies inside the other's circle; overlap between the circles alone is insufficient. The radius depends on the drive amplitude, so the diagram represents the peak-drive scale rather than a fixed interaction boundary throughout the pulse.

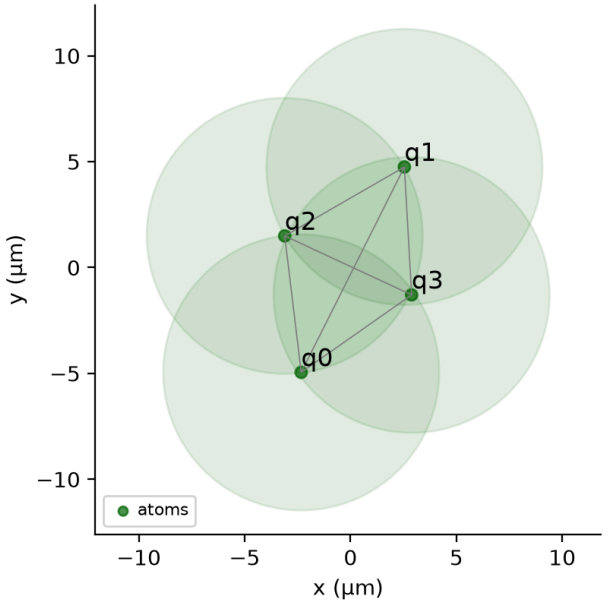


Figure 1. Embedded four-atom register for Tier-S instance seed 1. Circles have radius $R_b = 6.5161, \mu\mathrm{m}$, evaluated at $\Omega_p = 11.3097, \mathrm{rad}/\mu\mathrm{s}$. Drawn edges identify pairs separated by less than R_b and do not indicate vehicle membership. All six pair interactions remain in the Hamiltonian, including interactions beyond R_b.

The resulting global-detuning sequences are simulated using Pasqal's emu-sv state-vector emulator. For this example, the simulation uses a $10, \mathrm{ns}$ time step and 1,000 measurements, without pulse optimization or hardware noise modeling. Measured bitstrings are decoded into customer assignments and checked against the original capacity constraints before their clustering and routed costs are evaluated.

4. Software Architecture, Validation, and Evaluation

4.1. Instance Handling and Model Construction

The Python implementation separates instance generation, QUBO construction, classical references, quantum emulation, and evaluation [11]. Each run begins with a generated or saved instance containing customer coordinates, demands, depot position, capacity, and a feasible-assignment certificate. The instance parameters and certificate are validated before optimization.

NumPy represents the coordinates, distances, and symmetric QUBO matrix M. Model construction is independent of the solver, allowing the same objective to be examined through exhaustive enumeration and quantum emulation.

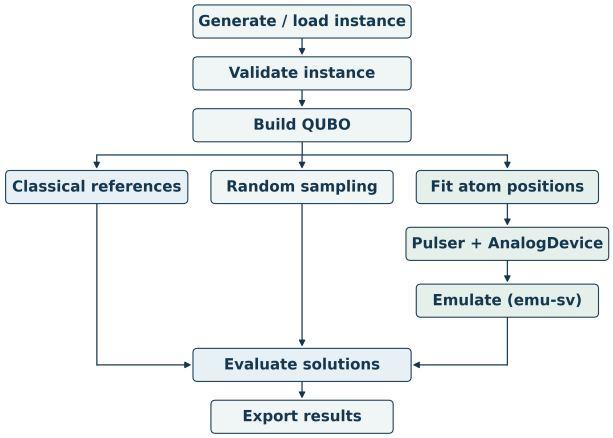


Figure 2. Software pipeline for Tier-S experiments. Classical references, uniform random sampling, and quantum emulation feed a shared evaluator, which checks feasibility and computes comparable solution-quality metrics.

4.2. Classical References and Validation

For Tier S, brute force evaluates all $2^4 = 16$ assignment bitstrings. It establishes the minimum original clustering cost among capacity-feasible assignments and, separately, the minimum penalized QUBO energy over all assignments. Comparing these optima tests whether the balancing surrogate preserves the desired clustering solutions.

The CVRP reference is obtained by considering feasible partitions and determining the shortest depot-to-depot route within each cluster. This distinguishes the optimum of the chosen clustering model from the optimum of the routing problem.

Additional enumeration checks the ground states of the target QUBO, the global-detuning approximation, and the embedded final Hamiltonian. These diagnostics record whether optimal clusterings survive each approximation and whether every resulting ground state remains feasible and clustering-optimal.

4.3. Pulser Sequence Construction and Emulation

SciPy's SLSQP optimizer fits physical atom coordinates to the target interactions. These coordinates define a Pulser *Register*. The register and *AnalogDevice* specification are combined in a Pulser *Sequence*, which contains the selected channel and pulse schedule.

AnalogDevice supplies the interaction coefficient and physical restrictions, including atom separation, register radius, drive amplitude, detuning, and sequence duration. Pulser checks the register and pulse operations against the selected device specification as the sequence is constructed [12]. Passing these checks establishes compatibility with that specification; it does not establish that the encoded Hamiltonian accurately represents the original objective.

The amplitude and detuning waveforms are applied through the global Rydberg channel. The completed sequence is passed to emu-sv, which numerically evolves the quantum state under the programmed Hamiltonian [13]. Thus, Pulser defines and validates the program, while emu-sv performs the state-vector simulation.

For the example in Section 3, emulation uses the CPU, a configured $10,\text{ns}$ time step, and 1,000 measurements. Final-state probabilities are retained separately from sampled counts, distinguishing the simulated distribution from finite-shot fluctuations.

4.4. Shared Evaluation and Reproducibility

Emulated and uniformly random bitstrings pass through the same evaluator. It decodes assignments, calculates vehicle loads, checks the original capacity constraints, and evaluates clustering costs. Feasible clusters are also routed exactly to obtain their CVRP costs.

Reported metrics include feasibility rate, best feasible model and CVRP gaps, and reference-reaching probabilities. Assignments differing only by vehicle labels are treated as equivalent clusterings. Gaps use the original objectives rather than penalized QUBO energies.

Benchmark outputs include CSV metrics and detailed JSON records containing instance data, seeds, settings, atom positions, encoded matrices, pulse parameters, measurement counts, and execution timings. These records allow aggregate results to be traced back to individual runs.

5. Tier-S Results

5.1. Benchmark Setup and Overall Performance

We evaluated 20 certified Tier-S instances, using seeds 0–19, with 1,000 measurements per instance for both emu-sv and uniform random sampling. The quantum sequences used four atoms, $\lambda = 0.1$, and fixed $4, \mu\text{s}$ pulses without pulse optimization or DMM. The benchmark records a $1, \text{ns}$ emulator time step. Exhaustive classical calculations established proven clustering and CVRP optima.

Figure 3 compares the mean per-instance probabilities. Emulation achieved a feasibility rate of 96.47%, compared with 58.66% for uniform random sampling. Its probability of reaching the clustering optimum was 63.97%, compared with 13.10%, while its CVRP-reference probability was 27.93%, compared with 12.76%. These reference-reaching probabilities are unconditional fractions of all measurements, rather than fractions restricted to feasible measurements.

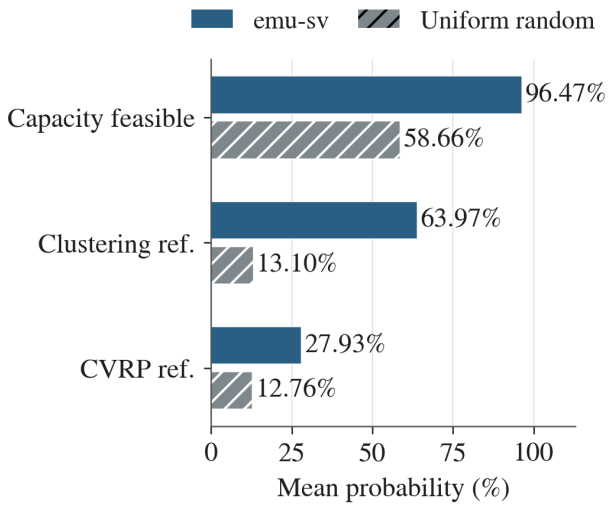


Figure 3. Mean per-instance feasibility and reference-reaching probabilities across 20 Tier-S instances. Each sampled method uses 1,000 measurements per instance. CVRP costs are obtained by routing feasible clusters exactly.

Both methods recovered a clustering-optimal assignment on every instance, giving a mean best model gap of 0%. The difference between their clustering-reference probabilities therefore concerns how frequently optimal assignments were sampled, rather than whether at least one was found within the measurement budget.

5.2. Variation Between Instances

The mean clustering-reference probability conceals substantial instance dependence. Figure 4 shows that most emulated instances concentrate probability near the clustering optimum, while several assign it very little probability. Uniform random sampling varies less between instances and generally has a lower success probability than the high-performing emulated cases.

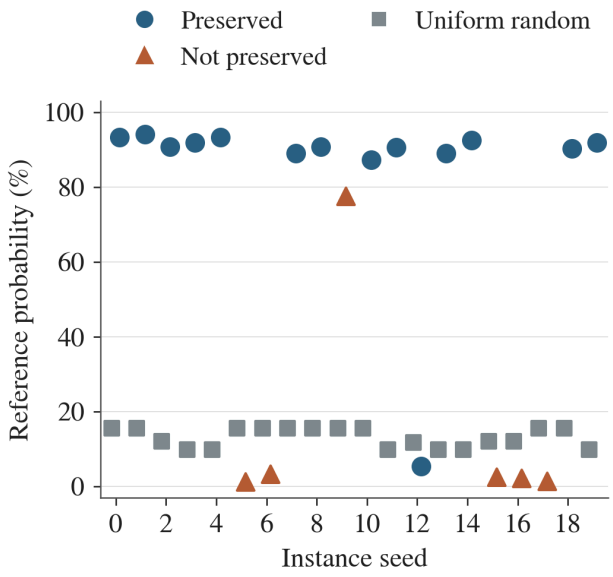


Figure 4. Measured clustering-reference probability by instance seed. Blue circles indicate that every embedded final-Hamiltonian ground state is clustering-optimal; orange triangles indicate that this condition fails. Gray squares show uniform random sampling.

Thus, the aggregate improvement does not hold uniformly across the benchmark. Some instances perform worse than the random baseline despite the high overall feasibility rate.

5.3. Ground-State Preservation

Figure 5 separates the effects of the surrogate formulation and the Hamiltonian approximations. The ideal QUBO had exclusively clustering-optimal ground states on 19 of 20 instances. After replacing the unequal diagonal coefficients with their mean global detuning, this count fell to 12. Following geometric embedding, it increased to 14.

The increase from 12 to 14 occurred on seeds 11 and 18. Consequently, geometric approximation does not necessarily degrade the ground-state ranking further: changes to the interactions can partially compensate for the preceding diagonal approximation.

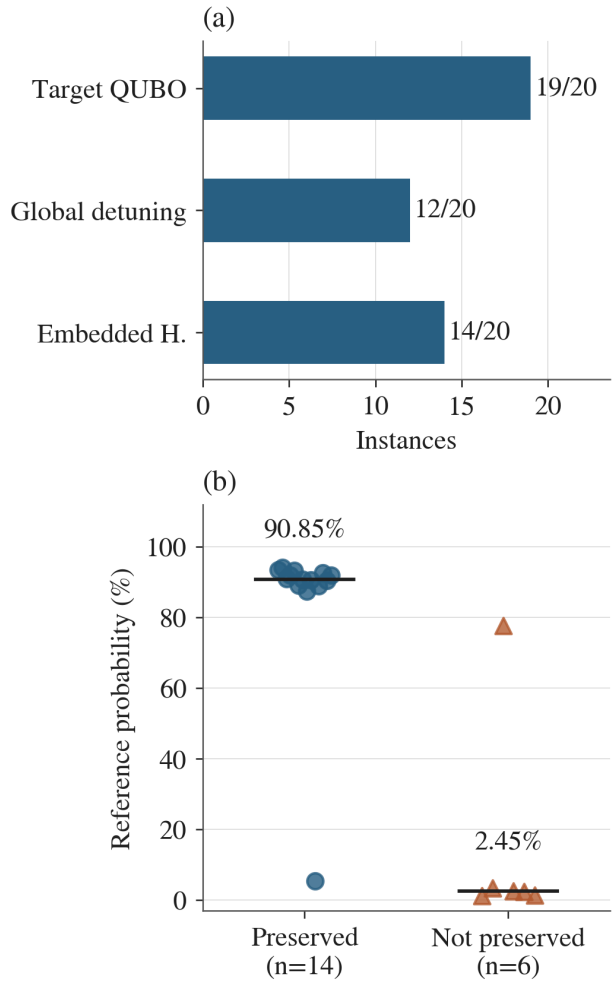


Figure 5. (a) Number of instances for which every ground state at each encoding stage is clustering-optimal. (b) Individual clustering-reference probabilities grouped by this property for the embedded Hamiltonian. Black lines indicate group medians.

The 14 instances satisfying the embedded ground-state criterion had a median clustering-reference probability of 90.85%, compared with 2.45% for the remaining six. Their mean feasibility rates were nevertheless nearly identical: 96.49% and 96.43%, respectively. Capacity feasibility alone therefore does not reveal the large difference in clustering quality.

Ground-state preservation is associated with sampling success but does not determine it. Seed 12 retained exclusively optimal ground states yet reached the clustering reference in only 5.4% of measurements. Conversely, seed 9 failed this criterion but achieved 77.6% success. These exceptions show why final-Hamiltonian diagnostics must be considered alongside finite-duration evolution.

The mean relative interaction error was 17.03%. Useful sampling therefore remained possible with imperfect interaction matching, but interaction fidelity and ground-state preservation measure different aspects of the encoding.

5.4. Routed CVRP Performance

Figure 6 shows the best feasible CVRP gap obtained within each measurement budget. Emulation recovered the CVRP optimum on 17 of 20 instances. The remaining gaps were approximately 41.2%, 59.7%, and 8.4% for seeds 7, 14, and 17, respectively, producing a mean best CVRP gap of 5.47%. Uniform random sampling recovered the CVRP optimum on all 20 instances.

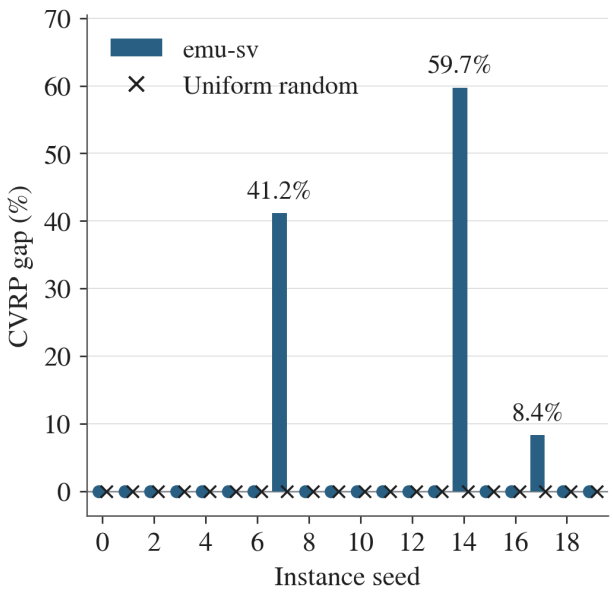


Figure 6. Best feasible CVRP gap from 1,000 measurements per instance. Bars show emulation results, with circles marking zero gaps; crosses show uniform random sampling. Each feasible partition is routed exactly before comparison with the proven CVRP optimum.

This outcome is compatible with emulation's higher mean per-shot CVRP-reference probability. A higher average probability does not ensure success on every instance, and uniform random sampling explores the small $2^4 = 16$-bitstring space extensively within 1,000 draws.

Furthermore, every emulated instance attained a 0% best clustering gap, including the three with nonzero CVRP gaps. Optimizing the clustering surrogate therefore does not guarantee the best routed solution. The results demonstrate improved mean per-shot feasibility and reference-reaching probabilities, but do not establish a computational advantage over classical sampling.

6. Scaling

6.1. Tier-M Formulation

Tier M contains $N = 10$ customers and $K = 3$ vehicles of capacity $Q = 20$. We extend the assignment model using two bits per customer, with the third vehicle represented implicitly [14] :

a_i	b_i	**Assignment**
0	0	Vehicle 3
1	0	Vehicle 1
0	1	Vehicle 2
1	1	Invalid

The assignment indicators are

$$x_{i1} = a_i, \qquad x_{i2} = b_i, \\ x_{i3} = 1 - a_i - b_i. \tag{35}$$

This construction eliminates one variable from the one-hot assignment equation $\sum_k x_{ik} = 1$. It remains valid provided $a_i + b_i \leq 1$. The invalid combination is penalized by

$$P_{\text{assignment}} = \eta \sum_i a_i b_i, \qquad \eta > 0. \tag{36}$$

This is the two-variable case of the standard pairwise at-most-one penalty [15]. It contributes zero for each valid code and η for 11.

Defining $L_k = \sum_i d_i x_{ik}$, the complete objective is

$$F_M(a,b) = \sum_{i<j} D_{ij} \sum_{k=1}^{3} x_{ik}x_{jk} + \lambda \sum_{k=1}^{3} \left(L_k - \frac{T}{3} \right)^2 + \eta \sum_i a_i b_i. \tag{37}$$

Because every indicator is affine in a_i, b_i, the objective remains quadratic. All pairwise coefficients are positive, and the encoding requires $2N = 20$ atoms, compared with 45 variables for conventional one-hot assignment with binary capacity slack.

Unlike the capacity surrogate, assignment validity can be guaranteed for the ideal QUBO. A sufficient bound is

$$\eta > \max_i(-h_i), \quad h_i = -\sum_{j \neq i} D_{ij} + 2\lambda(d_i^2 - T d_i). \tag{38}$$

For an invalid code, switching either active bit off removes its assignment penalty and cannot increase the other positive pair terms. Under this bound, the energy strictly decreases. This guarantee concerns ideal ground states; it does not guarantee that measured configurations are valid.

The implementation selects λ through an empirical rule based on total demand and mean pairwise distance [14]. For the supplied instance, $\lambda \approx 5.725$ and $\eta \approx 5060.509$. Enumeration of all $3^{10} = 59{,}049$ valid assignments, together with the assignment-validity bound, confirmed that every ideal QUBO ground state was clustering-optimal. Capacity nevertheless remains approximated rather than exactly encoded.

6.2. Embedding and Preliminary Emulation

The tested 20-atom layout satisfied the device's geometric limits: its minimum separation was $5.085, \mu\mathrm{m}$ and maximum register radius was $19.217, \mu\mathrm{m}$. The embedding optimizer reported convergence, but the relative interaction error remained 63.62%.

The supplied run used emu-mps with maximum bond dimension 64, precision setting 10^{-5}, a $20, \mathrm{ns}$ time step, and a $2, \mu\mathrm{s}$ pulse. DMM was enabled to accommodate unequal local detunings; these results therefore belong to the emulation-only DMM configuration.

Metric	Result
Relative interaction error	63.62%
Assignment-valid measurements	$85/200$ (42.5%)
Fully feasible measurements	$1/200$ (0.5%)
Clustering-reference hits	$0/200$
Best feasible model gap	44.85%

These results distinguish a device-legal layout from an accurate encoding. The register contains 190 pair interactions, all constrained by a common planar geometry. The conservative assignment penalty also introduces strong within-customer interactions. Rescaling the coefficients to hardware limits reduces the absolute energy scale of smaller objective differences, which can make resolving useful clusterings more demanding at a fixed pulse duration.

These are preliminary results for one instance and configuration. They do not isolate the contributions of the surrogate, geometric distortion, pulse dynamics, and numerical approximation.

6.3. Opportunities for Improvement

Further work could explore additional embedding restarts, joint optimization of coordinates and energy scale, and less conservative assignment penalties whose validity is still certified. Embedding objectives could also emphasize preserving low-energy solution rankings rather than interaction error alone.

Pulse amplitude, detuning trajectory, and duration remain candidates for optimization, following methods demonstrated for Rydberg-array optimization [10]. Such improvements require new benchmarks rather than an assumption that longer pulses will resolve the observed failures. Because emu-mps uses a matrix-product-state representation, bond-dimension and time-step convergence checks should accompany these experiments [16].

6.4. Tier L and Beyond

Tier L contains 15 customers and four vehicles. Extending the same construction directly—three explicit assignment indicators and one implicit vehicle—would require $N(K - 1) = 45$ variables before any capacity slack. This remains below the challenge's 80-atom limit, although embeddability and sampling performance would require separate validation.

The previously projected 30-variable count corresponds to two-bit binary vehicle labels. It is an assignment-information count, rather than a completed extension of the present quadratic formulation: label-equality and load expressions can introduce higher-order terms, requiring a separate derivation and potentially additional variables.

Beyond Tier L, useful scaling depends on the coupling structure, penalty-induced energy scales, available controls, and emulator/QPU convergence as well as atom count.

7. Conclusion

We developed and evaluated a compact, slack-free formulation for capacity-constrained clustering on analog neutral-atom registers. For Tier S, one binary variable per customer represents every assignment using four atoms. The resulting QUBO has positive pairwise coefficients compatible with repulsive Rydberg interactions, but replaces exact capacity encoding with a load-balancing surrogate. Variable efficiency therefore comes with a requirement to verify feasibility and objective preservation independently.

Across 20 Tier-S instances, global-detuning emulation improved mean per-shot feasibility and clustering-reference probabilities relative to uniform random sampling. However, the mean results conceal substantial instance dependence. Every embedded ground state was clustering-optimal on 14 instances; median clustering success was 90.85% for this group and 2.45% for the remaining six. Ground-state preservation is consequently an informative diagnostic, although finite-duration dynamics also affect sampling success.

The routing evaluation further limits the interpretation of clustering performance. Both sampled methods recovered a clustering optimum on every instance, yet emulation recovered the full CVRP optimum on only 17, compared with all 20 for uniform random sampling. Improving the clustering objective does not necessarily improve the best routed solution within a fixed measurement budget.

The preliminary Tier-M experiment demonstrates that atom count alone is insufficient to establish scalability. Although its 20-atom register satisfied geometric limits, substantial interaction distortion accompanied low assignment validity and capacity feasibility. Future work should jointly address penalty selection, embedding quality, and pulse design, with numerical convergence checks and broader instance benchmarks. The central result is a reduction in encoding overhead, together with evidence that useful neutral-atom optimization depends on preserving relevant solution rankings throughout formulation, embedding, and evolution.

References

[1] Pasqal, “Pasqal’s challenge: Analog quantum devices for optimization problems,” *Quantum Hackathon Challenge Brief*, ver. 3.0, unpublished challenge document, Oct. 24, 2026.

[2] S. Ebadi *et al*., “Quantum optimization of maximum independent set using Rydberg atom arrays,” *Science*, vol. 376, no. 6598, pp. 1209–1215, 2022, doi: 10.1126/science.abo6587.

[3] F. Glover, G. Kochenberger, and Y. Du, “A tutorial on formulating and using QUBO models,” arXiv:1811.11538, 2019. [Online]. Available: https://arxiv.org/abs/1811.11538

[4] A. Lucas, “Ising formulations of many NP problems,” *Frontiers in Physics*, vol. 2, Art. no. 5, 2014, doi: 10.3389/fphy.2014.00005.

[5] D-Wave, “Reformulating a problem,” *D-Wave Quantum Computing Products Documentation*. [Online]. Available: https://docs.dwavequantum.com/en/latest/quantum_research/reformulating.html. [Accessed: Oct. 25, 2026].

[6] A. Montañez-Barrera, D. Willsch, A. Maldonado-Romo, and K. Michielsen, “Unbalanced penalization: A new approach to encode inequality constraints of combinatorial problems for quantum optimization algorithms,” arXiv:2211.13914, 2022. [Online]. Available: https://arxiv.org/abs/2211.13914

[7] Pasqal, “QAA to solve a QUBO problem,” *Pulser Documentation*. [Online]. Available: https://pulser.readthedocs.io/en/stable/tutorials/qubo.html. [Accessed: Oct. 25, 2026].

[8] S. Lichtenberg and A. Tasissa, “A dual basis approach to multidimensional scaling,” *Linear Algebra and Its Applications*, vol. 682, pp. 86–95, 2024, doi: 10.1016/j.laa.2023.11.004.

[9] Pasqal, “Proportional-diagonal drive,” *Pasqal Documentation, QUBO Solver*. [Online]. Available: https://docs.pasqal.com/applicationsolvingtools/qubo/under_the_hood/driveshaping/proportional_diagonal/. [Accessed: Oct. 25, 2026].

[10] D. Turyansky, Y. Zolti, Y. Cohen, and A. Pick, “Pulse optimization in adiabatic quantum computation and control,” *Physical Review Research*, vol. 8, Art. no. 013206, 2026, doi: 10.1103/dq3h-1k35.

[11] 1Aiden-K, “Slack-Free Quantum QUBO Model,” *GitHub*. [Online]. Available: https://github.com/1Aiden-K/slack-free-quantum-qubo-mode. [Accessed: Oct. 26, 2026].

[12] Pasqal, “Sequence creation and validation,” *Pulser Documentation*, ver. 1.9.1. [Online]. Available: https://pulser.readthedocs.io/en/stable/sequence.html. [Accessed: Oct. 26, 2026].

[13] Pasqal, “Welcome to emu-sv,” *Pasqal Analog Emulators Documentation*. [Online]. Available: https://pasqal-io.github.io/emulators/latest/emu_sv/. [Accessed: Oct. 26, 2026].

[14] 1Aiden-K, “Slack-Free Quantum QUBO Model,” *GitHub*, tier-m-implementation branch. [Online]. Available: https://github.com/1Aiden-K/slack-free-quantum-qubo-model/tree/tier-m-implementation. [Accessed: Oct. 25, 2026].

[15] S. Takasugi, H. Nakaya, Y. Nakao, and K. Sawamura, “A smaller and more efficient one-hot encoding for QUBO,” *Scientific Reports*, vol. 16, Art. no. 25644, 2026, doi: 10.1038/s41598-026-55521-w.

[16] Pasqal, “Welcome to emu-mps,” *Pasqal Analog Emulators Documentation*. [Online]. Available: https://pasqal-io.github.io/emulators/latest/emu_mps/. [Accessed: Oct. 26, 2026].